

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and trained a **LightGBM classifier** on a dataset of ~1.1M transaction records to detect fraudulent activity, achieving an **AUC of 0.91** on held-out data.
- Engineered Python feature pipelines on **AWS Lambda** and **S3** to support downstream ML scoring, processing approximately **3,500 events/sec** during peak hours.
- Designed and implemented REST APIs using **Java Spring Boot** for inter-service data exchange, incorporating request validation and circuit-breaker patterns to enhance system resilience.
- Acted as a primary technical contact for three enterprise customers during their onboarding phase, gathering feature pipeline requirements and translating them into actionable **Python implementations**.
- Collaborated with business analysts and operations teams to gather requirements for a financial-services dashboard, focusing on **transaction monitoring** and operational alerting for failed payments.
- Wrote production runbooks and internal documentation that reduced customer onboarding time from over 4 weeks to approximately **10 days**.

SKILLS

Machine Learning – LightGBM, scikit-learn, gradient boosting, classification, AUC, F1-score, regression, supervised learning, unsupervised learning, evaluation metrics, ML Ops

Programming Languages – Python, Java, SQL, R

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux

Data Engineering & BI – Pandas, NumPy, Power BI, Tableau, DAX, Power Query, SQL (PostgreSQL, MySQL, Oracle), data warehousing, ETL, data modelling

Backend Development – Spring Boot, REST APIs, FastAPI

Development Tools & Methodologies – Git, Jenkins, Jira, Confluence, Agile, Scrum, CI/CD

PROJECTS

Fraud Detection Model on Transaction Features

- Engineered a feature pipeline using **Python** and **SQL** to extract behavioural signals from approximately 1.1M transaction records.
- Trained a **LightGBM classifier** on the generated features, tuning class weights to address a 1:140 fraud imbalance and achieving **0.91 AUC** and 0.46 F1.
- Packaged the trained model behind a **FastAPI** inference endpoint, implementing input validation and basic feature-drift logging with p95 latency around 180ms.

Customer Onboarding Playbook for ML Feature Store

- Worked directly with two pilot customers to design and deliver a guided onboarding flow for an internal ML feature store, including SDK examples and debug tooling.
- Built a **Python CLI** to validate customer feature definitions against the platform schema before submission, catching approximately **70%** of schema errors at write time.
- Developed a playbook and ran two follow-up workshops with customer data teams to upskill their internal engineers on the ML platform, reducing time-to-first-feature from ~6 weeks to 9 days.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)